DEPARTMENT ELECTIVE PROJECT · FINAL PRESENTATION

Aspect-Based Sentiment Analysis of Product Reviews

What we built, what the numbers say, and how a second training round (v2) improved on the first (v1).

TF-IDF · BERT · RoBERTa-base · RoBERTa-large · SemEval-2014 Laptops · MAMS

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HEADLINE RESULTS

79.9
macro-F1 on SemEval Laptops (RoBERTa-large v2)

85.1
macro-F1 on MAMS (RoBERTa-large v2)

84.6
aspect-extraction span F1 on Laptops (RoBERTa-base v2)

RECAP

One sentence, several opinions

ABSA splits the problem into two sub-tasks, and we evaluate every model in four train/test configurations.

*"The laptop has an excellent **display**, but the **battery life** is disappointing."*

SUB-TASK 1

Aspect Term Extraction

- Token-level BIO tagging
- the/O battery/B life/I is/O ...
- Scored by exact-match span precision, recall, F1

SUB-TASK 2

Aspect Sentiment Classification

- Input: sentence + one target aspect
- Output: positive / negative / neutral
- Scored by accuracy and macro-F1 (class imbalance)

PROTOCOL

Four configurations

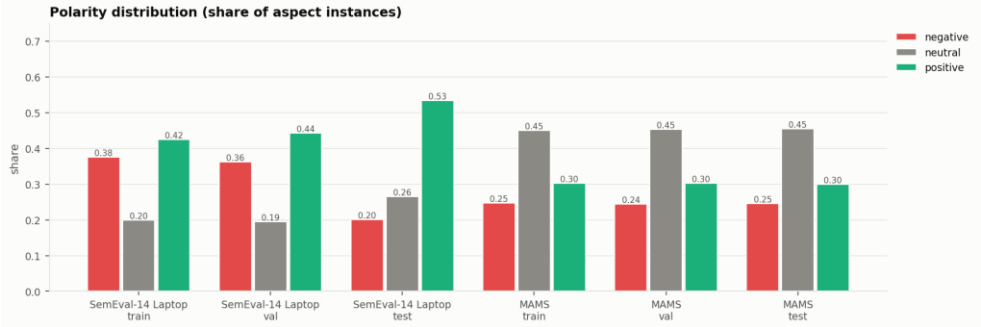
- Train Laptop → test Laptop
- Train MAMS → test MAMS
- Train Laptop → test MAMS
- Train MAMS → test Laptop
- Plus an end-to-end run: extract, then classify

DATA

Two corpora with very different shapes

SemEval-2014 Laptops (product reviews) and MAMS (restaurant reviews, built so every sentence mixes polarities).

Dataset	Split	Sentences	Aspects	Pos	Neg	Neu
SemEval Laptop	train	2,741	2,092	889	786	417
SemEval Laptop	val	304	221	98	80	43
SemEval Laptop	test	800	638	341	128	169
MAMS	train	4,297	11,186	3,380	2,764	5,042
MAMS	val	500	1,332	403	325	604
MAMS	test	500	1,336	400	329	607



11% / 100%

sentences with mixed polarities (Laptop / MAMS)

2.6 vs 1.6

aspects per sentence (MAMS vs Laptop)

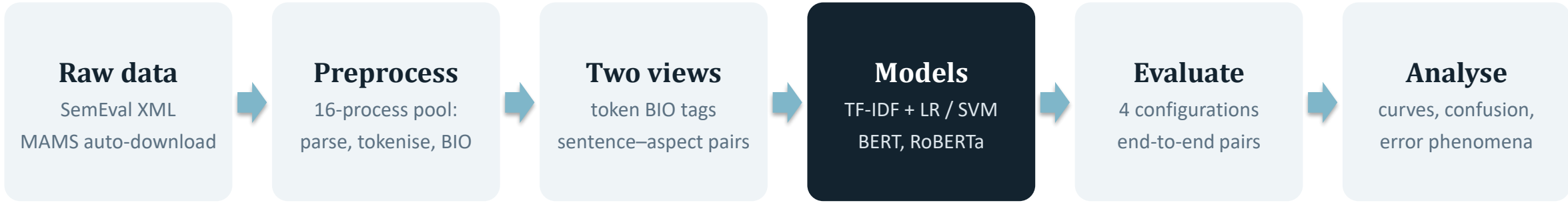
0.008

Jaccard overlap of aspect vocabularies (train)

WHAT WE BUILT

A reproducible pipeline, end to end

One command runs everything: python run_all.py (v1) and python run_v2.py (v2). Runs are cached and resumable.



ENGINEERING

Fast on a laptop GPU (RTX 4060, 8 GB)

- Whole tokenised dataset kept on the GPU, batches trimmed to their longest sentence
- bf16 mixed precision, TF32 matmuls, fused AdamW
- Preprocessing in ~8 s and cached; full v1 study in ~17 min

TRACKING

Every run leaves a trail

- Train and validation loss + metrics several times per epoch
- Early stopping on validation F1, best checkpoint restored
- Per-run curves, LR schedule, confusion matrices, live history.csv

ROUND 1 (V1)

Models compared under identical conditions

BASELINE

TF-IDF + LR / SVM

- Sentence 1–2-gram TF-IDF
- Context window: 6 words left/right of the aspect, side-tagged
- Aspect term TF-IDF
- C tuned on validation, grid run in parallel
- ATE: logistic-regression BIO tagger on word-window features

TRANSFORMERS

BERT-base & RoBERTa-base

- ASC input: [CLS] sentence [SEP] aspect [SEP]
- Classifier on the [CLS] vector
- ATE: token classifier, label on first sub-word
- AdamW, linear warm-up/decay, LR 2e-5 (ASC) / 5e-5 (ATE)
- 6 (ASC) / 8 (ATE) epochs, batch 32

EVALUATION

Same splits, same metrics

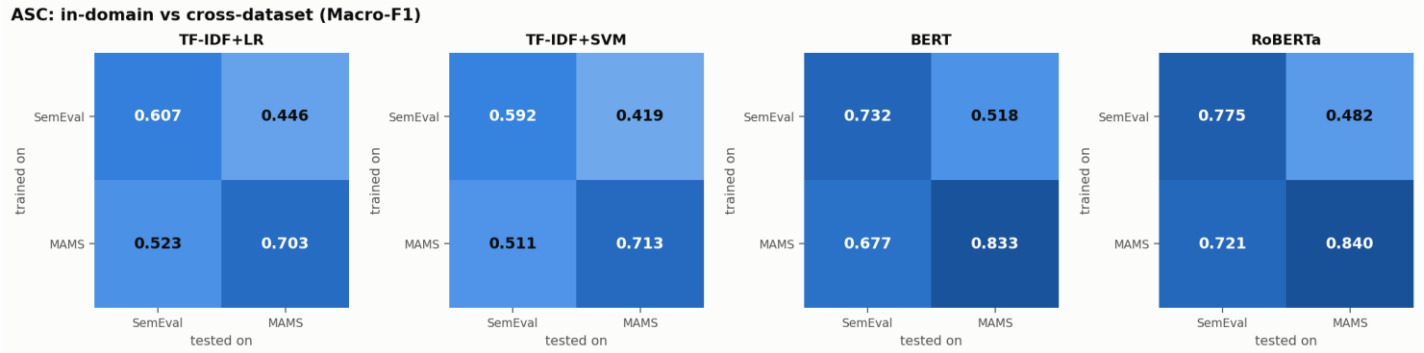
- ASC: accuracy, macro-P/R/F1, per-class F1
- ATE: exact-match span P/R/F1
- End-to-end: aspect span and polarity both right
- Checkpoint chosen on validation only; test touched once

ROUND 1 RESULTS

Transformers beat TF-IDF everywhere, but transfer is lopsided

Aspect sentiment classification, macro-F1 (%) on the test sets.

Model	Laptop → Laptop	MAMS → MAMS	Laptop → MAMS	MAMS → Laptop
TF-IDF+LR	60.7	70.3	44.6	52.3
TF-IDF+SVM	59.2	71.3	41.9	51.1
BERT-base v1	73.2	83.3	**51.8**	67.7
RoBERTa-base v1	**77.5**	**84.0**	48.2	**72.1**



READING THE TABLE

Three observations

- Contextual models add 17 macro-F1 points over TF-IDF on Laptops
- MAMS → Laptop keeps 72 F1, Laptop → MAMS only 48
- A model trained on mostly single-polarity sentences never learns to separate aspects that disagree
- On Laptops, neutral is the weakest class for every model

ROUND 1 RESULTS

Aspect extraction is domain-bound

Span F1 (%) for aspect term extraction and for the full extract-then-classify pipeline.

Aspect term extraction (span F1)

Model	Laptop → Laptop	MAMS → MAMS	Laptop → MAMS	MAMS → Laptop
TF-IDF+LR	57.8	68.6	19.6	11.1
BERT-base v1	81.4	75.0	24.2	31.6
RoBERTa-base v1	**85.9**	**76.2**	**44.0**	**39.5**

End-to-end (aspect and polarity both correct)

Model	Laptop → Laptop	MAMS → MAMS	Laptop → MAMS	MAMS → Laptop
TF-IDF+LR	41.4	52.1	10.8	6.5
BERT-base v1	63.5	64.2	14.2	24.6
RoBERTa-base v1	**69.1**	**65.5**	**27.6**	**29.9**

WHY

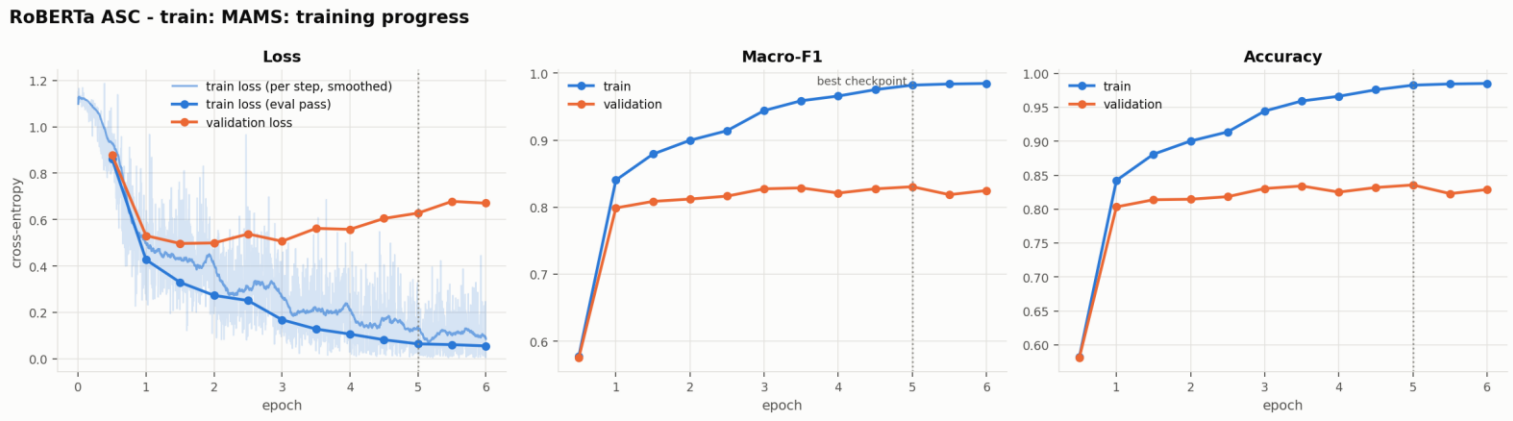
What limits extraction

- In-domain RoBERTa reaches the mid-80s on Laptops
- Laptop → MAMS: precision 76% but recall 31% — restaurant aspects are simply unseen
- End-to-end scores sit well below ASC alone: every missed aspect is also a missed sentiment

DIAGNOSIS

Why were the v1 numbers limited?

Reading the training curves and per-class scores pointed to four causes rather than one bug.



v1 RoBERTa-base, trained on MAMS

Blue = train, orange = validation. The gap opens after the first epoch and keeps widening.

1

Over-fitting

train macro-F1 reaches 98 while validation stalls near 83; validation loss rises after ~1.5 epochs

2

Aspect localisation

the [CLS] head is never told where the aspect sits — a problem when MAMS sentences carry 2–3 aspects with different labels

3

Neutral class

neutral F1 only 68 on Laptops; most errors are neutral ↔ positive/negative

4

Noisy selection

SemEval validation has 221 aspects, so validation F1 jumps ±3 points between checks

ROUND 2 (V2)

Each change answers one diagnosis

v1 problem	v2 change	Task
Over-fitting	Label smoothing 0.1 · FGM adversarial training on embeddings (ε = 0.5) · layer-wise LR decay	ASC + ATE
Aspect localisation	Aspect wrapped in the sentence: "… the # battery life # is …"; head reads [CLS] ⊕ mean of aspect tokens	ASC
Noisy selection	4 validation checks per epoch, batch 16, patience counted in epochs	ASC + ATE
Invalid BIO output	Viterbi decoding that forbids O → I transitions	ATE
Model capacity	RoBERTa-large (355M params) with gradient checkpointing to fit 8 GB	ASC + ATE

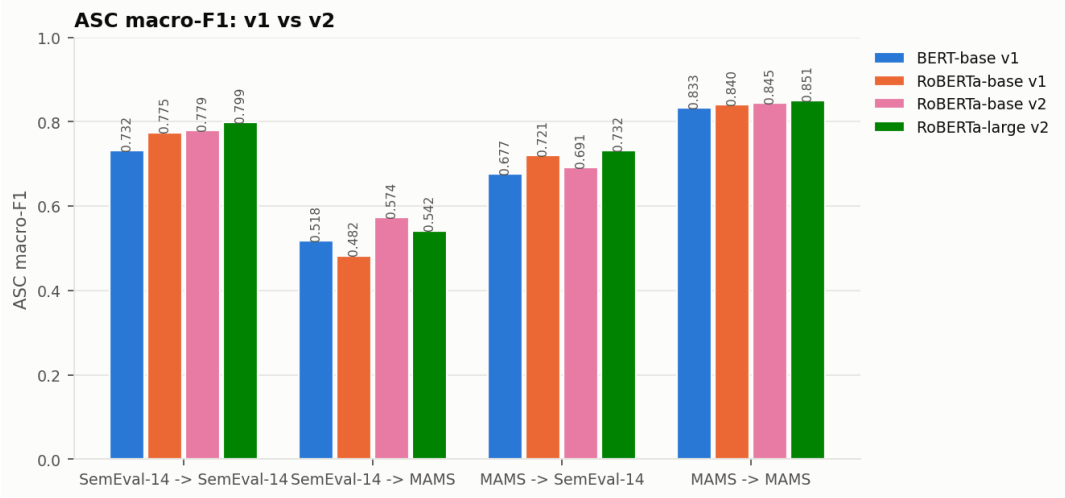
Controlled comparison: RoBERTa-base v2 uses the same backbone as v1, so its gain is due to the recipe alone. RoBERTa-large v2 adds model capacity on top.

Unchanged: data splits, metrics, test sets and the four configurations.

Aspect sentiment: where the second round helped

Macro-F1 (%) and change versus the best v1 model (RoBERTa-base v1).

Configuration	v1 base	v2 base	Δ	v2 large	Δ
Laptop → Laptop	77.5	77.9	**+0.4**	79.9	**+2.4**
MAMS → MAMS	84.0	84.5	**+0.5**	85.1	**+1.0**
Laptop → MAMS	48.2	57.4	**+9.2**	54.2	**+5.9**
MAMS → Laptop	72.1	69.1	**-3.0**	73.2	**+1.1**



Accuracy (%)

Configuration	v1 base	v2 large	Δ
Laptop → Laptop	80.4	83.2	**+2.8**
MAMS → MAMS	84.5	85.6	**+1.1**
Laptop → MAMS	48.7	55.6	**+6.9**
MAMS → Laptop	75.4	76.2	**+0.8**

V1 → V2

Extraction and the full pipeline

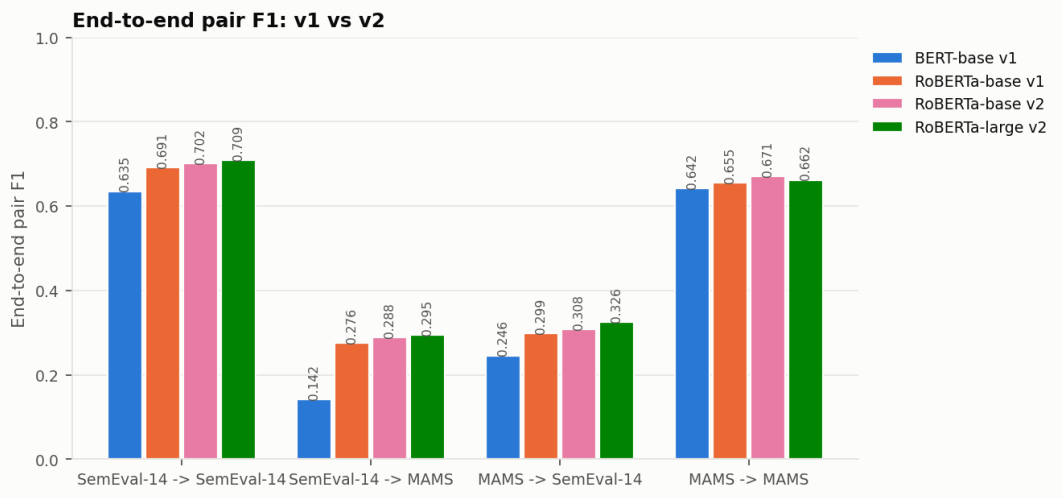
Span F1 (%) — change versus RoBERTa-base v1.

Aspect term extraction

Configuration	v1 base	v2 base	Δ	v2 large	Δ
Laptop → Laptop	85.9	84.6	-1.4	86.1	+0.2
MAMS → MAMS	76.2	78.0	+1.8	76.8	+0.6
Laptop → MAMS	44.0	38.3	-5.7	38.1	-5.9
MAMS → Laptop	39.5	41.5	+2.0	40.3	+0.8

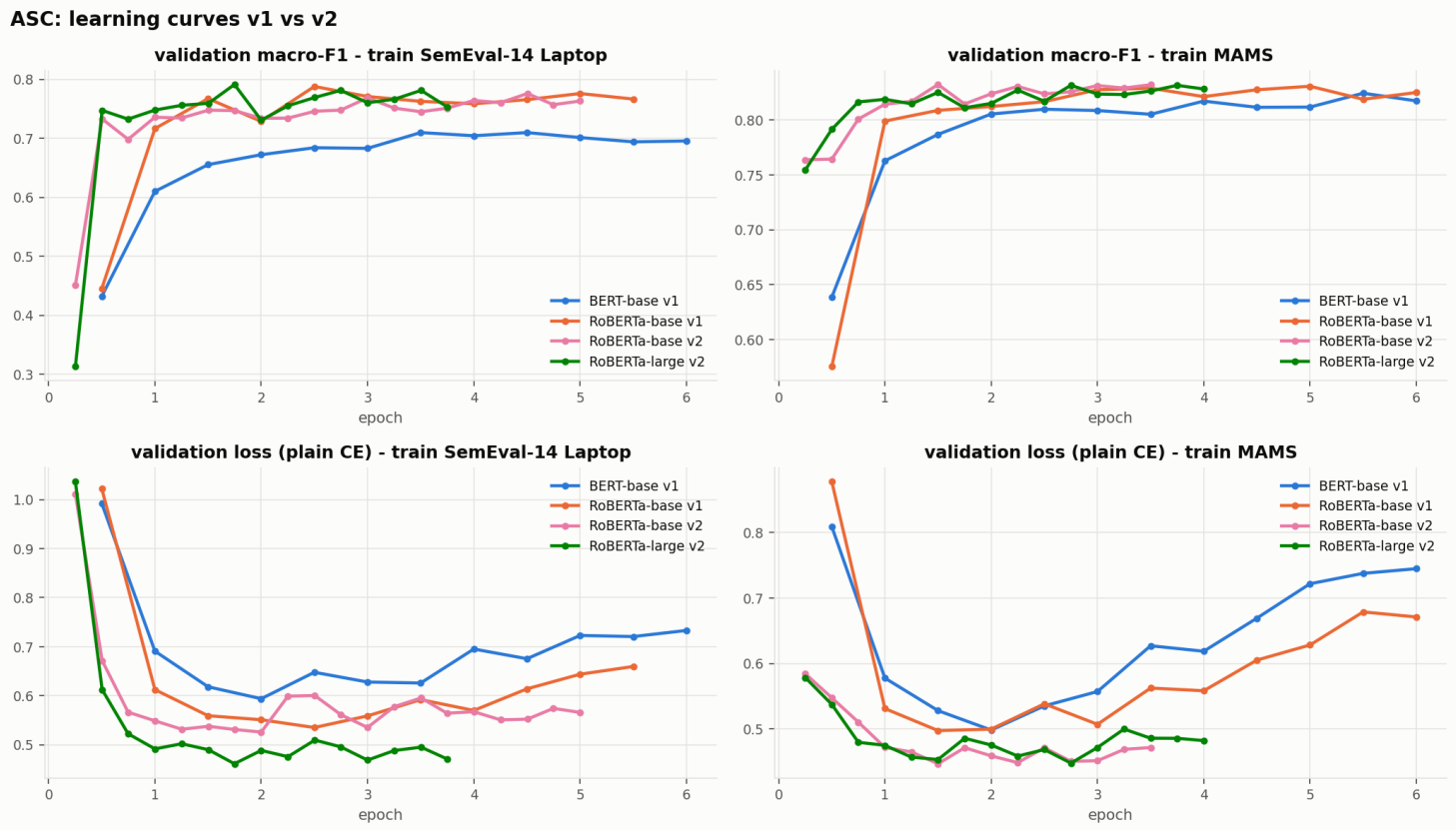
End-to-end pair F1

Configuration	v1 base	v2 base	Δ	v2 large	Δ
Laptop → Laptop	69.1	70.2	+1.0	70.9	+1.8
MAMS → MAMS	65.5	67.1	+1.6	66.2	+0.7
Laptop → MAMS	27.6	28.8	+1.3	29.5	+1.9
MAMS → Laptop	29.9	30.8	+0.9	32.6	+2.7



Learning dynamics: less over-fitting on MAMS, noisy on Laptops

Validation macro-F1 and validation loss per epoch (top: metric, bottom: plain cross-entropy).



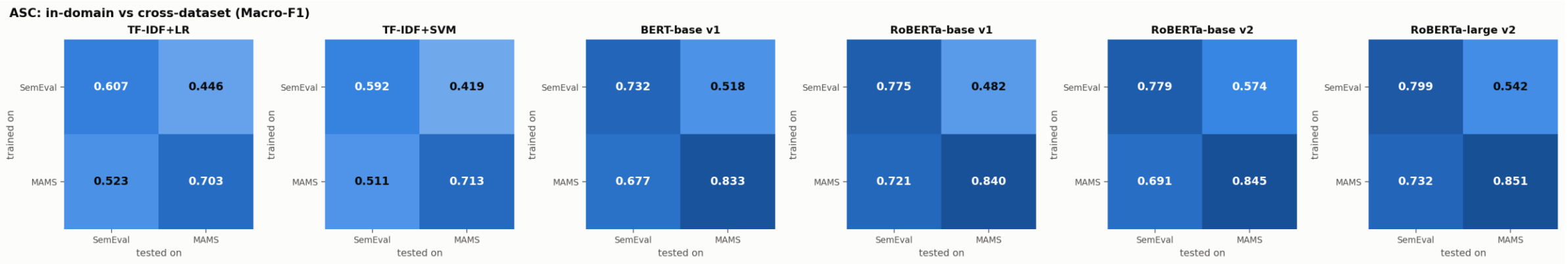
Gap at the selected checkpoint (ASC)

Model	Train	Train F1	Val F1	Gap
BERT-base v1	MAMS	98.8	82.4	16.3
RoBERTa-base v1	MAMS	98.2	83.1	15.2
RoBERTa-base v2	MAMS	91.0	83.2	7.8
RoBERTa-large v2	MAMS	96.0	83.1	12.9
BERT-base v1	Laptop	89.3	71.0	18.3
RoBERTa-base v1	Laptop	83.7	78.8	5.0
RoBERTa-base v2	Laptop	99.4	77.6	21.9
RoBERTa-large v2	Laptop	91.9	79.1	12.8

GENERALISATION

Cross-dataset transfer, all models

Rows: training corpus. Columns: test corpus. Macro-F1 for aspect sentiment classification.



TAKEAWAY

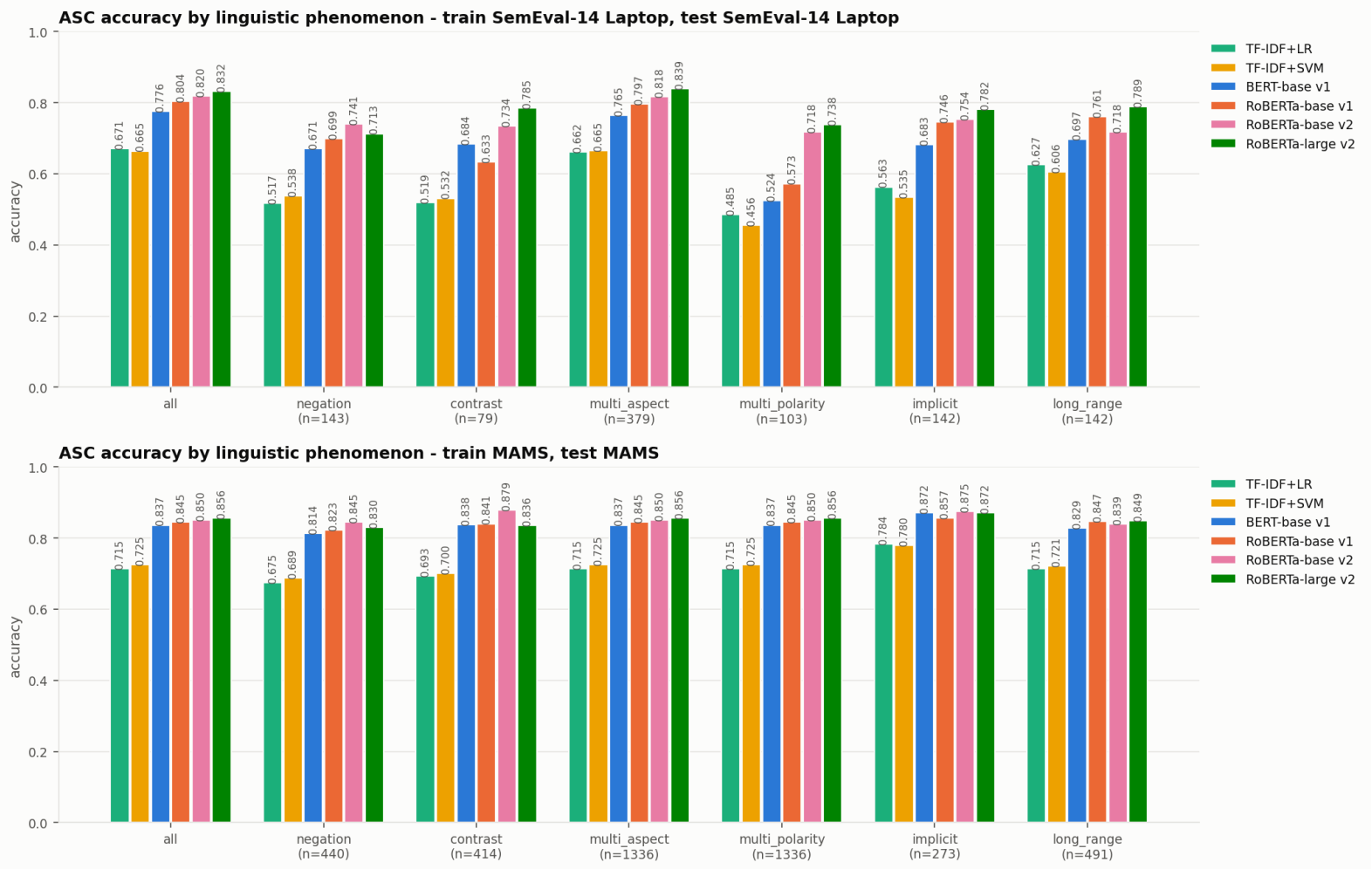
Transfer is asymmetric for every model family

- Best Laptop → MAMS: RoBERTa-base v2, 57.4 macro-F1
- Best MAMS → Laptop: RoBERTa-large v2, 73.2 macro-F1
- Training on multi-polarity text (MAMS) produces a model that also works on laptops; the reverse needs data, not tricks

ERROR ANALYSIS

Which sentences are still hard?

Accuracy of aspect sentiment by linguistic phenomenon, in-domain test sets.



BEST MODEL: ROBERTA-LARGE V2

What changed, what remains

- Mixed-polarity Laptop sentences: 57% (v1) → 74% (v2), the largest gain of any group
- Laptop: hardest = negation (71% vs 83% overall)
- MAMS: hardest = negation (83% vs 86% overall)
- Tags are rule-based: negators, contrast words, Liu's opinion lexicon, distance to nearest opinion word
- Misclassified examples per model are saved for inspection

DEMO

From raw text to aspect-level opinions

RoBERTa-base v2 extractor + classifier trained on Laptops, run on new sentences (notebook section 9).

Sentence	Aspect	Polarity	Conf.
The laptop has an excellent display, but the battery life is disappointing.	display	positive	0.95
The laptop has an excellent display, but the battery life is disappointing.	battery life	negative	0.93
The battery life is excellent but the keyboard is uncomfortable.	battery life	positive	0.94
The battery life is excellent but the keyboard is uncomfortable.	keyboard	negative	0.94
Customer support never answered my emails, although the screen is gorgeous.	Customer support	negative	0.94
Customer support never answered my emails, although the screen is gorgeous.	screen	positive	0.94
It boots up fast and the trackpad works fine, but the speakers are tinny.	boots up	positive	0.95
It boots up fast and the trackpad works fine, but the speakers are tinny.	trackpad	positive	0.94
It boots up fast and the trackpad works fine, but the speakers are tinny.	speakers	negative	0.95
I use it mostly for browsing and the price was okay.	price	positive	0.95

CONCLUSIONS

What we learned

FINDINGS

Results

- Fine-tuned Transformers beat TF-IDF by 14–17 macro-F1 points in-domain
- Best in-domain macro-F1: 79.9 (Laptop), 85.1 (MAMS)
- RoBERTa-large v2 beats the best v1 model in all 4 sentiment configurations; v2 improves end-to-end F1 everywhere
- Biggest v2 gain: mixed-polarity Laptop sentences, 57% → 74%
- Transfer depends on the shape of the training data more than on the model
- Aspect extraction does not survive a domain change

LIMITATIONS & NEXT STEPS

Honest caveats

- One random seed per run: differences of 1–2 points may not be significant
- v2 hurt Laptop → MAMS extraction (-5.7 pts) and base-v2 MAMS → Laptop sentiment (-3.0 pts)
- Small Laptop validation set (221 aspects) makes checkpoint choice noisy; the train–val gap grew there
- Phenomenon tags are heuristic, not hand-annotated
- Next: multi-seed averages, domain-adaptive pre-training on unlabeled reviews, joint ATE+ASC model, training on both corpora together

Thank you

Questions welcome

REPRODUCE

```
pip install -r requirements.txt
python run_all.py      # v1 study
python run_v2.py       # v2 study
ABSA_Notebook.ipynb    # results, figures and live demo
```

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